

Technical Monograph: Chlozolate (CAS 84332-86-5)

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Compound of Interest

Compound Name: Chlozolate

CAS No.: 72391-46-9

Cat. No.: B1210285

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Mechanisms, Toxicology, and Analytical Frameworks

Executive Summary

Chlozolate (Ethyl 3-(3,5-dichlorophenyl)-5-methyl-2,4-dioxo-1,3-oxazolidine-5-carboxylate) is a dicarboximide fungicide primarily utilized for the control of Botrytis, Sclerotinia, and Monilia species.^{[1][2][3]} While its agricultural utility is well-documented, its relevance to the scientific community extends into endocrine toxicology and osmotic signal transduction research.

This monograph synthesizes the compound's physicochemical properties with its molecular mechanism of action—specifically its interference with the fungal High Osmolarity Glycerol (HOG) pathway—and details the anti-androgenic toxicity profile that necessitates rigorous safety monitoring. Furthermore, we provide validated analytical frameworks for its quantification in complex matrices.

Physicochemical Profile

The lipophilic nature of **chlozolate** (LogP 3.15) dictates its environmental behavior and extraction protocols. It is rapidly hydrolyzed in alkaline conditions, a critical factor when designing stability studies or residue analysis workflows.

Property	Value	Relevance to Experimental Design
CAS Number	84332-86-5	Unique Identifier
Molecular Formula	C ₁₃ H ₁₁ Cl ₂ NO ₅	Mass Spectrometry precursor ion selection
Molecular Weight	332.14 g/mol	Calculation of molar dosage
Melting Point	112.6 °C	Solid-state characterization
Vapor Pressure	0.013 mPa (25°C)	Low volatility; suitable for GC with derivatization or direct injection
LogP (Kow)	3.15	High lipophilicity; requires organic solvents (e.g., Ethyl Acetate) for extraction
Solubility (Water)	~2 mg/L	Poor aqueous solubility; requires carrier solvents (DMSO/Acetone) for in vitro assays

Mechanism of Action: Osmotic Signal Transduction

Core Concept: Unlike sterol biosynthesis inhibitors (SBIs), **chlozolate** acts by disrupting the osmotic pressure regulation within fungal cells.

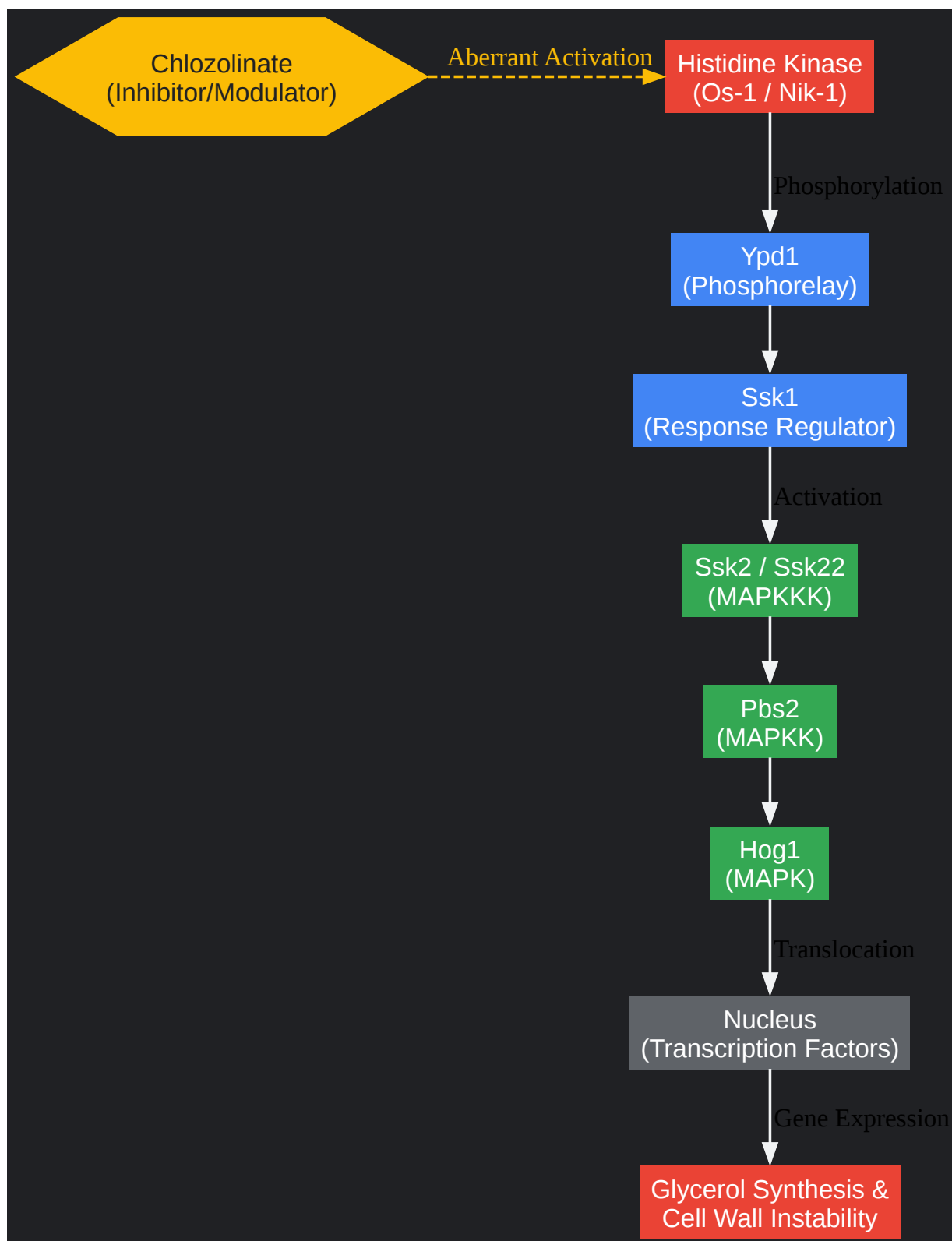
The Molecular Target: Histidine Kinase

Research indicates that dicarboximides, including **chlozolate**, target the Group III Histidine Kinase (HK) (orthologs of Os-1 in *Neurospora crassa* or Nik-1). This kinase is the upstream sensor for the High Osmolarity Glycerol (HOG) MAP kinase pathway.

- **Normal Function:** Under osmotic stress, the HK sensor phosphorylates the HOG pathway (Ypd1 → Ssk1 → Ssk2/22 → Pbs2 → Hog1), leading to glycerol synthesis and cellular adaptation.
- **Chlozolate Interference:** The fungicide induces "wrong" signaling, causing the hyperactivation of the HOG pathway even in the absence of osmotic stress. This results in excessive energy expenditure, glycerol accumulation, and cell wall instability, leading to fungal death.

Visualization: HOG Pathway Interference

The following diagram illustrates the signal transduction cascade and the point of dicarboximide intervention.



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Caption: **Chlozolate** induces aberrant signaling in the Histidine Kinase sensor, leading to lethal hyperactivation of the HOG MAPK cascade.

Toxicology & Endocrine Disruption

For drug development professionals, the toxicological profile of **chlozolate** serves as a model for anti-androgenic compounds.

Endocrine Disruption Mechanism

Chlozolate, like its structural analog vinclozolin, acts as an Androgen Receptor (AR) antagonist.

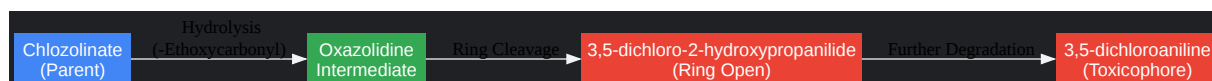
- Binding: Metabolites of **chlozolate** competitively bind to the AR, preventing the binding of endogenous androgens (testosterone/DHT).
- Consequence: In developmental toxicology studies, this blockade during the critical window of sexual differentiation leads to demasculinization of male offspring (e.g., reduced anogenital distance, hypospadias).

Metabolic Activation

It is critical to note that the parent compound often undergoes hydrolysis to active metabolites which may possess higher affinity for the AR than the parent.

Degradation Pathway:

- Hydrolysis: Loss of the ethoxycarbonyl group.
- Decarboxylation: Formation of 3-(3,5-dichlorophenyl)-5-methyloxazolidine-2,4-dione.[2]
- Ring Cleavage: Formation of 3,5-dichloroaniline derivatives (toxicophore).



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Caption: Metabolic degradation pathway of **Chlozolate** yielding potentially toxic anilide derivatives.

Analytical Methodologies

Precise quantification is required for residue monitoring and pharmacokinetic studies. The QuEChERS (Quick, Easy, Cheap, Effective, Rugged, and Safe) method followed by GC-MS/MS is the industry standard due to the compound's volatility and non-polar nature.

Protocol: QuEChERS Extraction & GC-MS/MS

Objective: Isolate **Chlozolate** from biological or plant matrices with a Limit of Quantification (LOQ) of 0.01 mg/kg.

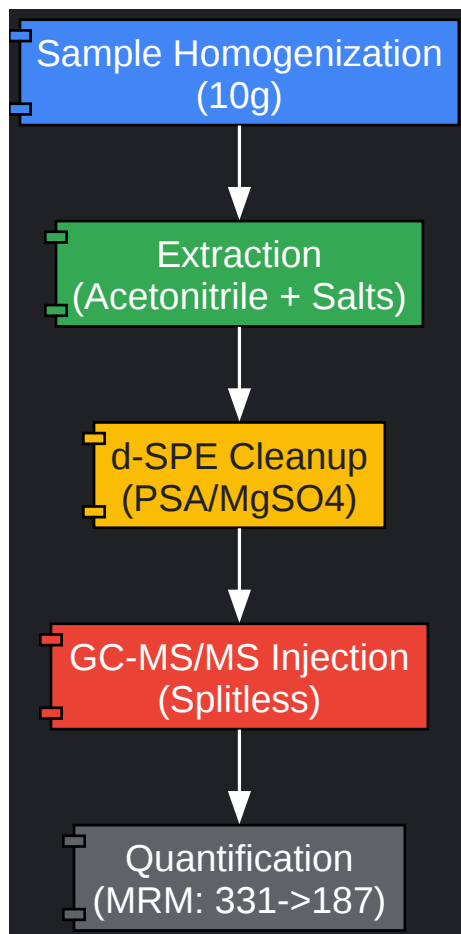
Step-by-Step Methodology:

- Homogenization: Weigh 10g of sample into a 50mL centrifuge tube.
- Extraction: Add 10mL Acetonitrile (or Ethyl Acetate for higher lipid content).
- Salting Out: Add 4g MgSO₄ and 1g NaCl. Vortex vigorously for 1 minute to induce phase separation.
- Centrifugation: Centrifuge at 3000 rpm for 5 minutes.
- Dispersive Solid Phase Extraction (d-SPE): Transfer an aliquot of the supernatant to a tube containing PSA (Primary Secondary Amine) to remove organic acids and sugars.
- Analysis: Inject 1μL into GC-MS/MS.

Instrument Parameters (Agilent 7010B Triple Quad recommended):

- Column: HP-5ms Ultra Inert (30m x 0.25mm x 0.25μm).
- Carrier Gas: Helium at 1.2 mL/min.
- Inlet Temp: 250°C (Splitless).
- MRM Transitions:

- Quantifier: 331 -> 187
- Qualifier: 331 -> 303



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Caption: Optimized QuEChERS extraction workflow for **Chlozolate** quantification.

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